

a. Examples:

- **Credit Risk Analysis:** Predicting loan defaults (low/high risk) for banks.
- **Email Spam Filtering:** Automatically classifying emails as spam or not spam.
- **Medical Diagnosis:** Identifying if medical images show disease or are healthy.
- **Sentiment Analysis:** Determining whether customer reviews are positive or negative.
- **Customer Churn Prediction:** Predicting if customers will stay or leave a service.

b. Classification Models

- **Logistic Regression**
 - **Why:** Simple, interpretable, fast training.
 - **Best for:** Binary classification tasks with linearly separable data (e.g., spam detection, credit approval).
- **Random Forest Classifier**
 - **Why:** Handles large datasets, reduces overfitting, good accuracy.
 - **Best for:** Complex, high-dimensional data (e.g., medical diagnosis, fraud detection).
- **Support Vector Machines (SVM)**
 - **Why:** Effective with clear decision boundaries, robust against overfitting.
 - **Best for:** Smaller datasets with clear margin between classes (e.g., image classification).
- **XGBoost / Gradient Boosted Trees**
 - **Why:** Highly accurate, robust, handles missing values and imbalanced datasets well.
 - **Best for:** Structured/tabular data with complex patterns (e.g., customer churn, predictive maintenance).
- **Neural Networks (Deep Learning)**
 - **Why:** Powerful in capturing complex, nonlinear relationships.
 - **Best for:** Large-scale datasets, particularly images, text, and speech (e.g., image recognition, sentiment analysis).

B- Regression

Regression – Predicting a **continuous numeric value** for each input.

a. Examples:

- **House Price Prediction:** Estimating property prices based on features like location and size.
- **Sales Forecasting:** Predicting future sales revenue for products or services.
- **Stock Price Prediction:** Estimating future stock prices using historical market data.